

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-a13c1601ce5b6958c.jsonl`  
- SHA-256: `b8086fa102dfc5c9c794172fe320d38a1ce6a93c779f1f4974ddc61cb4ece05a`  
- Source modified: `2026-05-31T04:23:37+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:59.234Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license), MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Pixtral-12B scores 90.7 ANLS on DocVQA and 81.8 (CoT) on ChartQA, indicating strong document-understanding/chart-reading capability relevant to parsing statement tables."

****Source:**** <https://huggingface.co/mistralai/Pixtral-12B-2409> (primary)

****Supporting quote:**** "DocVQA (ANLS): 90.7 ... ChartQA (CoT): 81.8"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:22:06.212Z)

I'll verify this claim about Pixtral-12B's benchmark scores. Let me start by checking the primary source and searching for corroborating/contradicting evidence.

3. assistant (2026-05-31T04:22:09.622Z)

I have the tools loaded. Let me verify the claim by checking the primary source and searching for corroborating evidence.

4. user (2026-05-31T04:22:16.010Z)

Benchmark Scores for DocVQA (ANLS) and ChartQA (CoT)

DocVQA (ANLS)

Model	Score
Pixtral 12B	90.7
Qwen2 7B VL	94.5
LLaVA-OV 7B	90.5
Phi-3 Vision	84.9
Phi-3.5 Vision	74.4
Claude-3 Haiku	74.6
Gemini-1.5 Flash 8B	79.5
LLaVA-OV 72B	91.6
GPT-4o	88.9
Claude-3.5 Sonnet	90.3

ChartQA (CoT)

Model	Score
Pixtral 12B	81.8
Qwen2 7B VL	38.6
LLaVA-OV 7B	67.1
Phi-3 Vision	72.0
Phi-3.5 Vision	67.7
Claude-3 Haiku	69.6
Gemini-1.5 Flash 8B	78.0
LLaVA-OV 72B	66.9
GPT-4o	85.1
Claude-3.5 Sonnet	87.6

Key Finding: Pixtral-12B-2409 achieves competitive performance on DocVQA (90.7) and particularly strong performance on ChartQA (81.8), outperforming many comparable models in its weight class.

5. user (2026-05-31T04:22:18.079Z)

Web search results for query: "Pixtral 12B DocVQA ANLS 90.7 ChartQA benchmark"

Links: [{"title": "Pixtral 12B 2409 · Models · Dataloop", "url": "https://dataloop.ai/library/model/mistralai_pixtral-12b-2409/"}, {"title": "Pixtral-12B vs Qwen3 VL 8B Thinking Comparison", "url": "https://llm-stats.com/models/compare/pixtral-12b-2409-vs-qwen3-vl-8b-thinking"}, {"title": "LLM Model: Pixtral-12B | DeepPranking.ai", "url": "https://deeppranking.ai/llm-models/pixtral-12b"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Pixtral-12B vs Qwen2.5 32B Instruct Comparison", "url": "https://llm-stats.com/models/compare/pixtral-12b-2409-vs-qwen-2.5-32b-instruct"}, {"title": "[Quick Review] Pixtral 12B", "url": "https://liner.com/review/pixtral-12b"}, {"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Pixtral 12B ? Another great multimodal model | UnfoldAI", "url": "https://unfoldai.com/pixtral-12b/"}]

Based on the search results, here's what I found about Pixtral 12B's performance on DocVQA and ChartQA:

Pixtral-12B-2409 achieves a DocVQA (ANLS) score of 90.7 and a ChartQA (CoT) score of 81.8. These are impressive benchmarks that demonstrate the model's strong performance on document and chart understanding tasks.

Pixtral 12B is a multimodal model with 12B parameters, plus a 400M parameter vision encoder, trained with interleaved image and text data, which means it can understand and respond to both images and text.

The DocVQA (Document Visual Question Answering) benchmark measures the model's ability to answer

questions about document images, while ChartQA tests its capability to interpret and answer questions about charts. Pixtral 12B's strong performance on DocVQA suggests its potential in automating document processing workflows, from contract analysis to form extraction.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:22:21.451Z)

The primary source (HuggingFace model card) confirms the exact numbers: DocVQA (ANLS) 90.7 and ChartQA (CoT) 81.8 for Pixtral 12B. The arxiv paper (Pixtral technical report) is also among the results. Let me verify against the primary academic source and check for any contradicting evidence or important caveats about these benchmark conditions.

7. user (2026-05-31T04:22:29.221Z)

Pixtral 12B Benchmark Results

Scores

According to Table 2 in the technical report:

- ****DocVQA (ANLS)****: 90.7
- ****ChartQA****: 81.8

Evaluation Methodology

Yes, the paper extensively discusses standardized evaluation practices. The authors state: "all models were evaluated with the same prompts, which are specified in Appendix A."

Key Caveats

The report identifies significant evaluation challenges:

1. **Prompt Sensitivity**: The paper shows that "commonly used prompts do not properly specify the output format," causing performance variations. They designed "Explicit" prompts that specify required output formats, finding that "leading models benefit greatly from 'Explicit' prompts."
2. **Metric Differences**: The authors found that exact-match metrics penalize substantively correct answers in different formats (e.g., "6.0" vs "6"). They introduce flexible parsing levels to account for this.
3. **Comparison Context**: For other models tested under this standardized harness, the paper notes discrepancies from reported figures. For instance, Llama-3.2 11B achieved only 14.8 on ChartQA under their protocol versus higher officially reported numbers.

The report emphasizes that "evaluation protocols for multimodal language models is not standardized," making their standardized approach notable for enabling fair cross-model comparison.

8. user (2026-05-31T04:22:31.358Z)

Web search results for query: "Pixtral 12B DocVQA ChartQA benchmark reproducibility dispute evaluation protocol"

Links: [{"title": "Papers Explained 219: Pixtral. Pixtral is a 12B parameter natively? | by Ritvik Rastogi | Medium", "url": "https://ritvik19.medium.com/papers-explained-219-pixtral-a714f94e59ac"}, {"title": "(PDF) Pixtral 12B", "url": "https://www.researchgate.net/publication/384770053_Pixtral_12B"}, {"title": "[Deprecated] Pixtral 12B | Mistral AI", "url": "https://mistral.ai/news/pixtral-12b/"}, {"title": "\\thetable Reproducing the reported performance of prior models. In Table we conduct fair re-evaluation of all models through the"}]

same evaluation harness, with the same prompt and metric. Here, we endeavour to recover the reported performance of all models by tuning evaluation settings towards individual models. We highlight that Pixtral 12B, like strong closed-source models (e.g. Gemini-1.5-Flash 8B [] and Claude-3 Haiku []) is able reports strong performance without such interventions.", "url": "https://arxiv.org/html/2410.07073v1"}, {"title": "[Quick Review] Pixtral 12B", "url": "https://liner.com/review/pixtral-12b"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Pixtral Large: Specs, Benchmarks & API Access Guide (2026)", "url": "https://ucstrategies.com/news/pixtral-large-specs-benchmarks-api-access-guide-2026/"}, {"title": "Pixtral 12B ? Another great multimodal model | UnfoldAI", "url": "https://unfoldai.com/pixtral-12b/"}]

Based on the search results, I found relevant information about Pixtral 12B and its evaluation on DocVQA and ChartQA benchmarks:

Evaluation Protocol and Reproducibility

The Pixtral 12B paper open-sourced the prompts used for evaluations, with prompts selected to reproduce reported performance of GPT-4o and Claude-3.5 Sonnet.

The evaluation re-evaluated all models through the same evaluation harness, with prompts designed for each dataset such that the results of leading multimodal models (GPT-4o and Claude-3.5 Sonnet) could be reproduced.

Benchmark Results

On ChartQA and DocVQA, Pixtral Large surpasses GPT-4o and Gemini-1.5 Pro. Pixtral 12B generally performs better with Explicit prompts, with only a minor regression on ChartQA.

Parsing Flexibility Analysis

Even with Explicit prompts, many models still provide outputs in various formats, which are penalized by metrics requiring responses to match reference answers exactly. To investigate this, models' generations are evaluated under progressively looser parsing constraints - for instance, flexible metrics do not penalize answers such as "6.0" or "The answer is 6" when the correct answer is "6".

The evaluation methodology involved re-evaluating all models under a common protocol with 'Explicit' prompts and analyzing sensitivity to parsing flexibility.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:22:39.767Z)

Both the primary HuggingFace model card and the arxiv technical report (Pixtral 12B, arxiv 2410.07073) confirm the exact numbers: DocVQA (ANLS) 90.7 and ChartQA (CoT) 81.8. These come from Mistral's own Table 2. Let me apply skeptical scrutiny to the framing of the claim.

The claim has two parts:

1. **Factual numbers**: DocVQA 90.7, ChartQA 81.8 ? confirmed by primary sources.
2. **Interpretive inference**: "indicating strong document-understanding/chart-reading capability relevant to parsing statement tables."

The key skeptical concerns to verify:

- These are **self-reported** vendor benchmarks (Mistral's own paper). The paper explicitly tunes prompts to reproduce GPT-4o/Claude numbers, and Pixtral benefits from its own "Explicit" prompt design. This is a known source of favorable-to-the-author bias.
- DocVQA/ChartQA measure **visual question answering** (answer a question about a doc/chart), NOT **full-table structured extraction** (transcribe every row of a dense financial table into JSON). High DocVQA does not establish reliability at row-level transaction extraction with

correct numbers.

Let me check whether DocVQA/ChartQA are good proxies for dense table extraction, and whether there's independent corroboration of Pixtral's table-extraction ability.

10. user (2026-05-31T04:22:48.303Z)

Web search results for query: "Pixtral 12B table extraction financial document accuracy independent evaluation OCR"

Links: [{"title": "Pixtral Large: Specs, Benchmarks & API Access Guide (2026)", "url": "https://ucstrategies.com/news/pixtral-large-specs-benchmarks-api-access-guide-2026/"}, {"title": "(PDF) Pixtral 12B", "url": "https://www.researchgate.net/publication/384770053_Pixtral_12B"}, {"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Arctic-Extract Technical Report", "url": "https://arxiv.org/html/2511.16470"}, {"title": "Pixtral-12B-2409 is now available on Amazon Bedrock Marketplace | Artificial Intelligence", "url": "https://aws.amazon.com/blogs/machine-learning/pixtral-12b-2409-is-now-available-on-amazon-bedrock-marketplace/"}, {"title": "Pixtral 12B: A Guide With Practical Examples | DataCamp", "url": "https://www.datacamp.com/tutorial/pixtral-12b"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Spatial ModernBERT: Spatial-Aware Transformer for Table and Key-Value Extraction in Financial Documents at Scale", "url": "https://arxiv.org/pdf/2507.08865"}]

Based on the search results, here's what I found about Pixtral 12B's table extraction and financial document accuracy:

Table Extraction Performance

Pixtral 12B achieved an 83.7% ChartQA score, demonstrating strong OCR and structured data extraction capabilities, and can parse complex charts, extract table data from screenshots, and reason about visual trends with acceptable accuracy.

Financial Document Application

The ChartQA performance suggests strong table extraction, which matters for financial documents where a single misread number breaks downstream analysis. The model is suitable for accurate extraction from visual documents including invoices, contracts, research papers with embedded charts, and financial reports with tables.

Comparative Evaluation

In an independent evaluation, Pixtral 12B achieved 55.23 ANLS (Approximate Matching in Layout and Semantic) on table extraction tasks, positioning it below several competing models including Llama 3.1 405B (80.37), Claude 4 Sonnet (79.65), and Qwen 2.5-VL (66.76).

Limitations

OCR accuracy drops significantly on non-Latin scripts like Chinese, Arabic, or Cyrillic, and the ChartQA benchmark only tested English. The benchmark scores are 18 months old as of March 2026, and Pixtral hasn't been re-evaluated on newer test sets.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. user (2026-05-31T04:22:48.541Z)

Web search results for query: "DocVQA benchmark single question answer not full table extraction limitation document AI"

Links: [{"title": "DocVQA Benchmark: 99.16% Accuracy Using Agentic Document Extraction - LandingAI", "url": "https://landing.ai/blog/superhuman-on-docvqa-without-images-in-qa-agentic-document-extraction"}, {"title": "BBox DocVQA: A Large Scale Bounding Box Grounded Dataset for Enhancing Reasoning in Document Visual Question Answer | alphaXiv", "url": "https://www.alphaxiv.org/overview/2511.15090v1"}, {"title": "DocVQA: A Dataset for VQA on Document Images", "url": "https://ukgovernmentbeis.github.io/inspect_evals/evals/multimodal/docvqa/"}, {"title": "DocVQA Benchmark Leaderboard", "url": "https://llm-stats.com/benchmarks/docvqa"}, {"title": "DocVQA: A Dataset for VQA on Document Images", "url": "https://arxiv.org/pdf/2007.00398"}, {"title": "BBox-DocVQA: A Large-Scale Bounding-Box?Grounded Dataset for Enhancing Reasoning in Document Visual Question Answer", "url": "https://arxiv.org/html/2511.15090v1"}, {"title": "M3DocRAG: Multi-modal Retrieval is What You Need for Multi-page Multi-document Understanding", "url": "https://arxiv.org/pdf/2411.04952"}]

Based on the search results, here's what I found about DocVQA benchmark limitations:

Key Limitations of DocVQA Benchmark

The DocVQA benchmark has a long-standing limitation of lacking fine-grained spatial grounding, with most existing DocVQA datasets limited to the page level.

Regarding the single question-answer focus, previous DocVQA datasets have questions that are specific to a single provided PDF (e.g., "What was the gross profit in the year 2009?"), rather than benchmarking open-domain question answering capabilities across multiple documents.

Additionally, even the strongest Vision Language Models exhibit significant failures in grounding?failures that directly impair their reasoning accuracy?demonstrating that current models are far from achieving robust spatial-semantic reasoning.

The research community has been addressing these limitations. For example, BBox DocVQA is a large-scale, bounding box grounded dataset designed to enhance spatial reasoning and evidence localization in visual documents, and M3DocRAG is a novel multi-modal RAG framework that flexibly accommodates various document contexts, question hops, and evidence modalities to overcome limitations of existing single-document DocVQA approaches.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-05-31T04:22:52.434Z)

This second search surfaces a critical data point. The Arctic-Extract Technical Report (arxiv 2511.16470) reports an **independent evaluation** where Pixtral 12B achieved only **55.23 ANLS** on table extraction tasks, far below the 90.7 DocVQA figure, and below competing models (Llama 3.1 405B 80.37, Claude 4 Sonnet 79.65, Qwen 2.5-VL 66.76). Let me verify this directly from the primary source.

13. user (2026-05-31T04:22:58.685Z)

Pixtral 12B Table Extraction Performance

Based on the report, Pixtral 12B's table extraction performance is:

Business Tables Dataset Score: 0.160 ± 0.02

This represents the lowest performance among all evaluated models on this benchmark. For comparison:

- Arctic-Extract: 0.720 ± 0.03
- Arctic-TILT: 0.725 ± 0.03
- Claude 4 Sonnet: 0.707 ± 0.03
- GPT5: 0.651 ± 0.04
- Qwen2.5-VL: 0.560 ± 0.03

The metric used is described as a specialized scoring system for transforming unstructured document information into structured tabular formats, though the specific metric name isn't explicitly stated for this particular benchmark. Pixtral 12B significantly underperformed all other models evaluated, suggesting substantial limitations in handling complex business table extraction tasks.

14. user (2026-05-31T04:23:02.059Z)

Web search results for query: "Pixtral 12B weak OCR dense text hallucination vision model criticism review"

Links: [{"title": "Pixtral 12B: Specs, Benchmarks & How to Deploy Mistral's Vision Model (2026)", "url": "https://ucstrategies.com/news/pixtral-12b-specs-benchmarks-how-to-deploy-mistrals-vision-model-2026/"}, {"title": "Pixtral 12B Model Review: Great for Images, Not So Much for Multilingual", "url": "https://www.yeschat.ai/blog-pixtral-12b-model-review-great-for-images-not-so-much-for-multilingual-53209"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/pdf/2410.07073"}, {"title": "[Deprecated] Pixtral 12B | Mistral AI", "url": "https://mistral.ai/news/pixtral-12b/"}, {"title": "Pixtral 12B", "url": "https://arxiv.org/html/2410.07073v2"}, {"title": "Vision Language Models Map Logos to Text via Semantic Entanglement in the Visual Projector", "url": "https://arxiv.org/pdf/2510.12287"}, {"title": "ReadBench: Measuring the Dense Text Visual Reading Ability of Vision-Language Models", "url": "https://arxiv.org/html/2505.19091"}]

Based on the search results, here's what I found about Pixtral 12B's weaknesses and criticisms:

Key Criticisms and Weaknesses

****Dense Text and Hallucination Issues:****

Vision hallucinations on dense scenes are the most common failure mode, with the model miscounting objects, inventing text in low-resolution regions, and struggling with spatial reasoning in cluttered images. User reports from late 2024 mention hallucinations on dense diagrams and unreliable object counting, with one user reporting the model counted 47 people in a crowd photo when the actual count was 31.

****OCR Limitations:****

The vision encoder sometimes invents text in low-resolution screenshots. Additionally, Pixtral 12B experiences notable degradation when processing dense text, even at shorter content lengths.

****Overall Limitations:****

The 12B scale means complex reasoning tasks fall apart, and it hallucinates object counts in dense images. The model trails larger models by 20 to 30 points on complex reasoning benchmarks and struggles with multi-step logic and advanced math.

****Maintenance Status:****

It's worth noting that Pixtral 12B is no longer maintained and has been replaced by more powerful vision and multimodal models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

15. assistant (2026-05-31T04:23:20.472Z)

I now have strong evidence to evaluate this claim carefully. Let me consolidate my findings.

****On the factual numbers (the quote):****

- The HuggingFace model card and the Mistral arxiv paper (2410.07073, Table 2) both confirm: DocVQA (ANLS) ****90.7**** and ChartQA (CoT) ****81.8**** for Pixtral 12B. The quote is accurate as a transcription of the source. ?

****On the inference ("indicating strong document-understanding/chart-reading capability relevant to parsing statement tables"):** This is where the claim overreaches, and three independent lines of evidence cut against it:

1. ****Self-reported vendor benchmarks with prompt-tuning.**** These are Mistral's own numbers from their own technical report. The paper explicitly tunes prompts and parsing flexibility, and Pixtral benefits from its self-designed "Explicit" prompts. Notably, under Mistral's harness competitors crater (Llama-3.2 11B got 14.8 on ChartQA vs. much higher official numbers), a red flag that the harness is favorable to Pixtral. Self-reported, prompt-tuned vendor benchmarks are exactly the "cherry-picked benchmark" pattern the checklist warns about.

2. ****DocVQA/ChartQA are poor proxies for the muneem task.**** DocVQA measures answering a *single question* about a document ("What was the gross profit in 2009?"); ChartQA answers questions about a chart. Neither measures *full multi-row structured table transcription* (extract every transaction row with correct date/narration/debit/credit/balance). Independent research confirms DocVQA lacks fine-grained grounding and even strong VLMs fail at spatial-semantic reasoning. A high VQA score does not establish reliability on dense financial-table extraction.

3. ****Independent evaluation directly contradicts the implied table capability.**** The Arctic-Extract Technical Report (arxiv 2511.16470) evaluated Pixtral 12B on a Business Tables extraction benchmark and it scored ****0.160 ? dead last**** among all models tested (vs Qwen2.5-VL 0.560, Claude 4 Sonnet 0.707, Arctic-Extract 0.720). User/independent reports also document Pixtral hallucinating/inventing text on dense, low-resolution regions and degrading on dense text ? the exact failure mode for dense Indian bank statements and a serious hallucination risk on financial numbers.

****Verdict:**** The raw numbers in the supporting quote are correctly transcribed and verifiable in the primary source ? so it is not a misread. But the claim's load-bearing assertion is the *inference* that these scores indicate "capability relevant to parsing statement tables." That inference is refuted: (a) the scores are self-reported vendor benchmarks under a self-favorable prompt-tuned harness (marketing-flavored, the paper is the vendor's own); (b) DocVQA/ChartQA do not measure full-table structured extraction, the actual muneem task; and (c) an independent benchmark shows Pixtral 12B is the ***worst*** model tested at business-table extraction (0.16). For a research report whose purpose is ranking models for *dense bank-statement table parsing*, citing 90.7/81.8 as evidence of relevant capability is misleading. The claim should be killed as framed.

Per the rubric (refuted if "marketing fluff / cherry-picked benchmark" OR the inference is contradicted), this meets multiple refutation conditions.

16. user (2026-05-31T04:23:31.735Z)

Structured output provided successfully

17. assistant (2026-05-31T04:23:37.570Z)

Verdict delivered: ****refuted = true**** (high confidence).

The supporting quote's raw numbers are accurate (DocVQA 90.7, ChartQA 81.8 confirmed in both the HuggingFace card and Mistral's own arxiv paper Table 2), so it is not a transcription error. But the claim is refuted on its load-bearing inference that these scores indicate capability "relevant to parsing statement tables":

1. ****Self-reported / cherry-picked****: the numbers come from Mistral's own paper under a self-favorable prompt-tuned harness where competitors implausibly crater (Llama-3.2 11B = 14.8 on ChartQA).
2. ****Wrong proxy****: DocVQA/ChartQA measure single-question VQA, not full multi-row table transcription ? muneem's actual task.
3. ****Directly contradicted****: the independent Arctic-Extract report (arxiv 2511.16470) puts Pixtral 12B ***dead last*** at business-table extraction (0.160 vs Qwen2.5-VL 0.560, Claude 4 Sonnet 0.707), and independent reports document it inventing text in dense/low-res regions ? the exact hallucination risk for dense Indian bank statements.